Accessibility' may be the 'holy grail' for the twentyfirst century city.

Public transport offers the greatest potential to enhance accessibility in cities, but is non-existent or declining in most developing countries, and increases in developed countries are not commensurate with the scale required to meet sustain ability targets.

Moving from captive riders (passengers) to clients and choice riders, making public transport a lifestyle choice, requires a strong customer focus.

Despite the multiple benefits it generates for both users and society as a whole, NMT is often marginalized and receives minimal priority in urban mobility planning and investments, both in developed and developing countries. It constitutes the principal and often only accessible means of transport for the majority of residents in developing-country cities with most who opt for this mode doing so out of a lack of choice. Yet, in most cities, NMT conditions are extremely hostile. Investing in NMT to enhance the safety and security of walking and cycling constitutes a key pillar of planning and design for accessible mobility in cities. Innovative experiences from both developed and developing-country cities that have elevated NMT as a foundation for urban sustainability offer valuable insights to inform planning and investments elsewhere.
In the absence of accessible public transport services, informal transport remains predominant in developing countries and constitutes the main means of motorized trips for most urban dwellers. Although it provides essential benefits to the urban poor in terms of mobility and livelihoods, informal transport generates a number of environmental and economic externalities. A significant challenge is to balance the efficiency and social equity aspects of informal transport, i.e. to achieve the social benefits of free market services without exceeding social costs. The sector would benefit immensely from best-practice examples of successful regulation of service quality and safety while at the same time allowing the inherent advantages of private competition and entrepreneurship to flourish.
Given the current state of urban transport globally, improved urban accessibility requires focusing on a number of vital pillars. Increasing the modal share of ***public transport*** is a universally applicable strategy that has significant potential to address mobility challenges of both developed and developing countries.
Metro, light rail and BRT are all intended to provide fast, comfortable and cost-effective urban mobility in medium- to high demand corridors Metro systems require huge investments and are often implemented as the preferred option of large cities where demand justifies that high capital cost.
Although metro systems are the most expensive urban public transport option, their high capacity and best performance (in terms of speed and number of passengers conveyed), make them invaluable parts of highly developed trans port systems.

The term ‘BRT’ was initially coined in the US15 and the first wide-scale development of BRT was implemented in Curitiba, Brazil, in 1982. Other names for BRT are ‘high-capacity bus system’, ‘high-quality bus system’, ‘metro-bus’, ‘surface metro’, ‘express bus system’ and ‘busway system’.While the terms may vary from country to country, the basic premise is followed: a high-quality customer-oriented public transport that is fast, safe, comfortable, reliable and cost-effective. The best BRT systems flexibly combine stations, bus services, busways and information technologies into an integrated system with a strong identity. Depending on the specific system design, BRT capital costs are 4–20 times lower than light rail systems, and 10–100 times lower than metro systems, with similar capacity and service level.

Despite their growth, high-capacity public transport systems still face a number of challenges, especially in developing countries.

Integration occurs at three levels: physical, operational and fare. Physical integration allows for direct connections from one service to another, usually including transfer facilities and terminals. Operational integration consists of coordination of schedules and frequencies so that the service is guaranteed and wait times are not excessive. Fare integration involves free or reduced cost transfers, usually through advanced ticketing systems. Adequate integration requires the development of information systems to coordinate services and provide information to the users.

Integration with other elements of the transport system

Besides the integration between components of the public transport system, it is important to provide adequate connectivity with other components of the urban transport system, such as walking, biking, taxis, informal transport services, cars and motor -cycles. These types of connections complement public transport systems, as feeder services, to provide door-to-door connectivity and allow for expanded coverage of the public transport system.

Walking is usually the most common access mode to public transport and requires an adequate environment, with protected, well-lit, signalized and surfaced sidewalks. Design should consider the needs of the most vulnerable users: children, the elderly and people with disabilities. It is important to build these spaces according to good practices, but perhaps even more important is to keep such spaces clean and free of encroachments. Whereas the management of sidewalks is often outside the jurisdiction of public transport agencies, adequate coordination with the responsible agencies is important to ensure safe and pleasant travel for public transport passengers who are walking to and from the stations.

At important integration points, especially in the periphery of cities, adequate space is needed for these mechanisms. This is to ensure that different types of users are able to connect to the public transport system and avoid using cars to go to the city centre.

Integration with the built environment

Accompanied by complementary land-use and zoning policies, high-capacity public transport systems can encourage compact, pedestrian and public-trans port friendly environments that are integrated into the surrounding area. Several cities, such as Copenhagen (Denmark), Singapore and Curitiba (Brazil), have been able to implement efficient public trans port services and develop urban forms that are highly conducive to public trans port ridership. In these cities, public transport and urban form function in harmony: either through mixed-use, compact and accessible development suited for public transport (also known as transit-oriented development), or through flexible public transport options suited to low-density urban development.

Singapore is planned as a public-transport oriented compact city, with high-density residential and commercial developments around transport nodes. This improves accessibility to public transport.

Although public bus and train services are provided on a commercial basis, all forms of public transport are generally affordable to the public – thus contributing to increased use of public transport and a reduction in the use of private vehicles. The adequate integration between public transport and the built environment makes both the public transport system and the city successful. High density (combined with Disincentives to private car ownership and use) increases ridership, while public transport provides access to dense, accessible, mixed-use urban environments. Consequently, shorter trips can be completed on foot or bicycle. The result is less vehicle kilometres travelled and thus lower transport emissions and fewer traffic accidents.

Quality of service

Quality of service involves several elements as perceived by the user involving dimensions such as travel time, reliability, safety and security, comfort and user information. Travel time includes the door to door connectivity, walking to the station, waiting for the service, travelling on board, transferring between services and walking to the final destination.

Reliability involves the confidence on the arrival of the service, and the travel time on board.

Safety implies the buses and trains are well maintained and that passengers would not be exposed to preventable accidents. Security implies that passengers travel with the realization that they would not be victims of crime or terrorist attacks. Comfort deals with several amenities, but mainly with the space available, or occupancy. User information comes in many forms to allow the passenger to navigate the system and be aware of real time information and contingencies. The most advanced public transport systems in the world include all these dimensions of quality to provide a very attractive alternative to car and motorcycle use. Many advanced systems in developing countries have high-quality services, but may

not include the first and last leg of the trip (i.e. walking to and from the station). ‘Universal design’– which is an important aspect of inclusive public transport systems – is often overlooked.
Cities in developed countries have incorporated reliability as part of the key performance indicator metrics. Nevertheless, in developing cities, reliability is not commonly measured and hence not managed.

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The availability of finance is essential for efficient urban mobility systems. These systems play important social, economic and environmental roles in terms of facilitating more efficient urban mobility systems and sustainable urban devel opment patterns. Such high-capacity public transport systems are primarily appropriate for large and dense urban agglomerations, and serve as important parts of integrated public transport systems. Accordingly, they should be designed to provide a competitive and viable alternative to private cars and motorcycles. Integration is important for public transport systems to be efficient and sustainable. The most efficient systems are those that have achieved route integration; integration with other public trans port systems; integration with private motorized transport (including through encouraging drivers of private cars to park outside the city centre and use public trans port for parts of their daily commute); integration with non-motorized modes (through easy access for pedestrians and/or bicycle parking and allowing bicycles onto public trans port vehicles); and fare integration: allowing users to travel throughout the urban public trans port system on a single ticket, or at reduced rates when switching between operators and/or lines. Integration also includes the built environment dimension: dense, mixed-use and accessible urban forms enhance rider - ship and vice versa.
Heightened concerns over climate change, rising gasoline prices, traffic congestion and social exclusion have sparked renewed interest to explore the link between mobility and urban form. Worldwide, city officials share relatively similar concerns about travel time, air quality, road accidents, social integration, better accessibility and improved use of different modal transport solutions.
Despite this, most cities, particularly in developing countries and emerging economies, continue to prioritize motorized transport and related urban infrastructure. A large number of cities both in the developing and developed countries are experiencing fast and uncontrolled growth in their peripheries.
Consequently, there is a wide variety of urban forms, defined by land-use and transportation systems that are not conducive to the provision of ‘efficient’ forms of urban mobility. There can be little doubt that designing neighbour hoods, cities

and regions in a way that can reduce private car dependency, promote healthier, more sustainable urban forms and a variety of travel solutions, can make the city more accessible to all. The pressure to develop sustain able transportation and mobility systems is particularly acute in urban areas.

There is increasing evidence that the form and functionality of the city is crucial for the promotion of sustainable mobility. Indeed, transforming cities wherein a mix of activities is closer together, in a more compact configuration, and interlaced by high-quality pedestrian and bicycle infra structure, is Tantamount to the creation of a more *accessible* city.

As stressed in earlier, accessibility lies at the core of achieving an urban form that is environmentally sustainable, socially equitable and inclusive, with higher potential to generate economic interactions that lead to productivity and income gains. Sustainable mobility is an outcome of how cities and neighbour hoods are designed and take form, but it also shapes the urban form itself. This reflects the powerful, bi-directional relationship between mobility and urban form that underscores the importance of carefully coordinating and integrating the two. Density is but one element of urban form that influences travel. The spatial distribution of population and employment densities are also important.

Where people live, work, shop and socialize, sets the stage for travel by defining the location of trip origins and destination, and thus the length of trips and the energy they consume.

Many studies have been conducted on the potential to reduce motorized travel through changes in the built environment. Analysts often express features of built environ ments along five core dimensions or the ‘5 Ds’: density, diversity, design, destination accessibility and distance to transit. These 5 Ds strongly influence travel demand – notably, the number of trips made, the modes chosen and the distances travelled.